

Representation theory

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1. Derived algebraic geometry

1.1. Quasi-coherent and Ind-coherent sheaves

1.1.1. Categorical constructions

Definition 1.1.1.1. Throughout this note, the word category will mean $(\infty, 1)$ -categories unless stated otherwise. A functor is said to be continuous if it preserves all filtered colimits. Most importantly, we use homological indexing.

Definition 1.1.1.2. We fix a ground field k . We let $\mathbf{Vect} = \mathbf{Mod}_{Hk}(\mathbf{Sp})$ denote the category of module spectra of the Eilenberg-MacLane spectrum Hk regarded as a $\mathbb{E}_\infty$ -ring spectrum.

Definition 1.1.1.3. Notice that $\mathcal{D}^-(k)$ is a symmetric monoidal category equipped with the derived tensor. We define the category $\mathbf{DGCat}$ to be $\mathbf{Mod}_{\mathbf{Vect}}(\mathbf{Pr}_{\mathbf{St}}^L)$.

todo 1.1.1.1. Write something more here, at least explain left complete t -structure and blah blah.

1.1.2. The category of (pre)stacks

Definition 1.1.2.1. We define the category of affine schemes by $\mathbf{Aff} = \mathbf{CAlg}(\mathbf{Vect}_{\geq 0})^{\mathrm{op}}$.

Definition 1.1.2.2. We define the category of prestacks by $\mathbf{pStk} = \mathbf{pSh}(\mathbf{Aff}) = \mathbf{Fun}(\mathbf{Aff}^{\mathrm{op}}, \mathbf{Ani})$. By Yoneda embedding, we obtain an inclusion $\mathbf{Aff} \hookrightarrow \mathbf{pStk}$.

Definition 1.1.2.3. We now introduce a truncated version of affine schemes. Consider the subcategory of $\mathbf{Vect}$ given by the objects concentrated between degrees $[0, n]$, say $\mathbf{Vect}_{[0, n]}$. We have a symmetric monoidal structure on this subcategory given by $(A, B) \mapsto \tau_{\leq n}(A \otimes B)$, thus we may define the category of n -truncated affine schemes $\mathbf{Aff}_{\leq n} = \mathbf{CAlg}(\mathbf{Vect}_{[0, n]})^{\mathrm{op}}$. This category of commutative algebras is denoted by $\mathbf{CAlg}_{[0, n]}$ also for short.

Construction 1.1.2.4. Notice that the inclusion $\mathbf{Vect}_{[0, n]} \hookrightarrow \mathbf{Vect}_{\geq 0}$ is lax symmetric monoidal, and in particular preserves commutative algebras. Thus we have a fully faithful functor

$$\mathbf{CAlg}_{[0, n]} \hookrightarrow \mathbf{CAlg}_{\geq 0}$$

hence an inclusion $\mathbf{Aff}_{\leq n} \hookrightarrow \mathbf{Aff}$.

Notice that we have a truncation functor $\tau_{\leq n} : \mathbf{Vect}_{\geq 0} \rightarrow \mathbf{Vect}_{[0, n]}$, which naturally restricts to the commutative algebra category, that is, we have a commutative diagram

$$\begin{array}{ccc}
\mathbf{CAlg}_{\geq 0} & \xrightarrow{\tau_{\leq n}} & \mathbf{CAlg}_{[0,n]} \\
U \downarrow & & \downarrow U \\
\mathbf{Vect}_{\geq 0} & \xrightarrow{\tau_{\leq n}} & \mathbf{Vect}_{[0,n]}
\end{array}$$

where U is the forgetful functor. Moreover, $\tau_{\leq n}$ is left adjoint to the inclusion functor. We denote the functor induced on the category of affine schemes by taking op also by $\tau_{\leq n} : \mathbf{Aff} \rightarrow \mathbf{Aff}_{\leq n}$.

remark 1.1.2.1. We have $\mathbf{Aff}_{\leq 0}$ the classical category of affine schemes.

Definition 1.1.2.5. We define the category

$$\mathbf{Aff}_{<\infty} = \bigcup \mathbf{Aff}_{\leq n}$$

the category of eventually truncated affine schemes.

Definition 1.1.2.6. We define the category of n -prestacks by ${}_{\leq n}\mathbf{pStk} = \mathbf{pSh}(\mathbf{Aff}_{\leq n}^{\text{op}})$. Restriction along the inclusion $\mathbf{Aff}_{\leq n} \hookrightarrow \mathbf{Aff}$ gives a functor ${}_{\leq n}\tau : \mathbf{pStk} \rightarrow {}_{\leq n}\mathbf{pStk}$. Moreover, the left Kan extension functor gives a fully faithful left adjoint of ${}_{\leq n}\tau$, say $\text{LKE} : {}_{\leq n}\mathbf{pStk} \hookrightarrow \mathbf{pStk}$, which somehow justifies the name of the above truncation functor.

remark 1.1.2.2. Right Kan extension gives a right adjoint of the functor ${}_{\leq n}\tau$.

Warning 1.1.2.7. We should think ${}_{\leq n}\mathbf{pStk}$ as the prestacks n -truncated in the sense that they are detected or determined by all the maps from n -truncated affines, rather than the usual sense of truncatedness of presheaves, i.e., has values in n -truncated anima. We will introduce the latter notion $\mathbf{pStk}_{\leq n}$ later.

Definition 1.1.2.8. Let $\mathcal{X}$ be a prestack. We say $\mathcal{X}$ is convergent (or nilcomplete in the sense of [Lur18]) if for any $S \in \mathbf{Aff}$, we have $\mathcal{X}(S) \rightarrow \lim_n \mathcal{X}(\tau_{\leq n}(S))$ is an equivalence.

Lemma 1.1.2.9. Any prestack representable by an affine scheme is convergent.

Proof: By completeness of the t -structure in $\mathbf{Sp}$ hence in $\mathbf{Vect}$, for any algebra $A \in \mathbf{CAlg}(\mathbf{Vect}_{\geq 0})$ we have $A \rightarrow \lim \tau_{\leq n} A$ is an isomorphism. The rest follows by Yoneda embedding and the universal property of $\tau_{\leq n}$ as a left adjoint. $\square$

Example 1.1.2.3. The stack of isomorphisms of quasi-coherent sheaves is not convergent.

Proposition 1.1.2.10. A prestack $\mathcal{X}$ is convergent iff as a functor $\mathcal{X} : (\mathbf{Aff})^{\text{op}} \rightarrow \mathbf{Ani}$, we have the equivalence $\mathcal{X} \simeq \text{RKE}(\mathcal{X}|_{\mathbf{Aff}_{<\infty}})$

Proof: We first directly compute the right Kan extension of $\mathcal{Z} : (\mathbf{Aff}_{<\infty})^{\text{op}} \rightarrow \mathbf{Ani}$. By the point-wise limit formula,

$$\mathcal{Z}(S) = \lim_{S' \in (\mathbf{Aff}_{<\infty})_{/S}} \mathcal{Z}(S')$$

So it suffices to see that the functor $\mathbb{N} \rightarrow (\mathbf{Aff}_{<\infty})_{/S}$ by sending n to $\tau_{\leq n} S$ is cofinal in the category. We omit the higher bullshit here and verify only that it is cofinal on the level of underlying 1-categories. By definition, any object $T \in \mathbf{Aff}_{<\infty}$ is n -truncated for some n , and a map $T \rightarrow S$ hence factors through $T \rightarrow \tau_{\leq n} S$ by adjunction. $\square$

Construction 1.1.2.11. By the above proposition, if we let $\mathbf{pStk}_{\text{conv}} \subset \mathbf{pStk}$ denote the subcategory of convergent prestacks, this inclusion admit a left adjoint $\mathcal{X} \mapsto \mathcal{X}_{\text{conv}}$, constructed as $\mathcal{X}_{\text{conv}} = \text{RKE}(\mathcal{X}|_{\mathbf{Aff}_{<\infty}})$

1.1.3. Finite type conditions

Definition 1.1.3.1. We say an object $A \in \mathbf{CAlg}_{[0,\infty)}$ is of finite type if $H_0(A)$ is of finite type over k and $H_i(A)$ is finitely generated module over $H_0(A)$. We denote this category by $\mathbf{CAlg}_{[0,\infty)}^{\text{ft}}$. We denote the intersection $\mathbf{CAlg}_{[0,\infty)}^{\text{ft}} \cap \mathbf{CAlg}_{[0,n]} = \mathbf{CAlg}_{[0,n]}^{\text{ft}}$. Likewise on its opposite, we define $\mathbf{Aff}_{<\infty}^{\text{ft}}$ and $\mathbf{Aff}_{<\infty}^{\text{ft}} \cap \mathbf{Aff}_{\leq n} = \mathbf{Aff}_{\leq n}^{\text{ft}}$.

Theorem 1.1.3.2 ([Lur17]). The objects of $\mathbf{CAlg}_{[0,n]}^{\text{ft}}$ are compact in $\mathbf{CAlg}_{[0,n]}$. Moreover, for any $S \in \mathbf{Aff}_{\leq n}$, the category opposite to $I = (\mathbf{Aff}_{\leq n}^{\text{ft}})_{/S}$ is filtered, and the cofiltered limit $S \mapsto \lim_{S_0 \in I} S_0$ is an isomorphism.

Corollary 1.1.3.3. $\mathbf{CAlg}_{[0,n]}^{\text{ft}}$ is a subcategory of compact generators in $\mathbf{CAlg}_{[0,n]}$. Hence, $\mathbf{CAlg}_{[0,n]} = \text{Ind}(\mathbf{CAlg}_{[0,n]}^{\text{ft}})$. Dually, we have $\mathbf{Aff}_{\leq n} = \text{Pro}(\mathbf{Aff}_{\leq n}^{\text{ft}})$. Moreover, since $\mathbf{CAlg}_{[0,n]}^{\text{ft}}$ is idempotent complete, we have $\mathbf{CAlg}_{[0,n]}^{\text{ft}} = (\mathbf{CAlg}_{[0,n]})^{\omega}$.

Definition 1.1.3.4. Let $\mathcal{X} \in {}_{\leq n}\mathbf{pStk}$ for some n , we say it is locally of finite type if it is the left Kan extension $\mathcal{X} = \text{LKE}(\mathcal{X}|_{\mathbf{Aff}_{\leq n}^{\text{ft}}})$. We denote the full subcategory by ${}_{\leq n}\mathbf{pStk}^{\text{lft}}$.

Construction 1.1.3.5. Regarding the above definition as ${}_{\leq n}\mathbf{pStk}^{\text{lft}} = \text{pSh}(\mathbf{Aff}_{\leq n}^{\text{ft}})$, we may restate the above definition as there is an inclusion functor given by left Kan extension ${}_{\leq n}\mathbf{pStk}^{\text{lft}} \hookrightarrow {}_{\leq n}\mathbf{pStk}$ with a right adjoint given by restriction of sites.

Corollary 1.1.3.6. By [Theorem 1.1.3.2](#), a stack $\mathcal{X} \in {}_{\leq n}\mathbf{pStk}$, equivalent to a functor $\mathbf{CAlg}_{[0,n]} \rightarrow \mathbf{Ani}$ is left Kan extended from $\mathbf{CAlg}_{[0,n]}^{\text{ft}} = \mathbf{CAlg}_{[0,n]}^{\omega}$ iff it preserves filtered colimits. Thus, by expanding definitions, $\mathbf{Aff}_{\leq n} \cap {}_{\leq n}\mathbf{pStk}^{\text{lft}} = \mathbf{Aff}_{\leq n}^{\text{ft}}$.

Definition 1.1.3.7. We say an affine scheme $S \in \mathbf{Aff}$ is almost of finite type if for all n we have $\tau_{\leq n} S$ is of finite type. We denote the full subcategory of almost of finite type affine schemes by $\mathbf{Aff}_{\leq n}^{\text{aft}}$.

Dually, an algebra $A \in \mathbf{CAlg}_{\geq 0}$ is almost of finite type if $H_0(A)$ is of finite type over k and $H_n(A)$ is finitely generated as an $H_0(A)$ -module. We denote the category by $\mathbf{CAlg}_{\geq 0}^{\text{aft}}$.

Definition 1.1.3.8. We say a prestack $\mathcal{X} \in \mathbf{pStk}$ is locally almost of finite type if it is convergent and for every n , we have $\tau_{\leq n} \mathcal{X} \in \tau_{\leq n} \mathbf{pStk}^{\text{lft}}$. We denote the corresponding full subcategory by $\mathbf{pStk}^{\text{lft}}$.

Corollary 1.1.3.9. By [Corollary 1.1.3.6](#), $\mathbf{pStk}^{\text{lft}} \cap \mathbf{Aff} = \mathbf{Aff}^{\text{aft}}$.

Proposition 1.1.3.10. We have an equivalence of categories

$$\mathbf{pStk}^{\text{lft}} = \mathbf{pSh}(\mathbf{Aff}_{< \infty}^{\text{ft}})$$

where $\mathbf{pStk}^{\text{lft}} \rightarrow \mathbf{pSh}(\mathbf{Aff}_{< \infty}^{\text{ft}})$ is given by restricting along the inclusion $\mathbf{Aff}_{< \infty}^{\text{ft}} \hookrightarrow \mathbf{Aff}$, and the inverse is given by $\mathbf{RKE}_{\mathbf{Aff}_{< \infty} \hookrightarrow \mathbf{Aff}} \mathbf{LKE}_{\mathbf{Aff}_{< \infty}^{\text{ft}} \hookrightarrow \mathbf{Aff}_{< \infty}}$.

Proof: By [Proposition 1.1.2.10](#), since we require a locally almost of finite type prestack $\mathcal{X}$ to be convergent, it suffices to identify the following two conditions on a convergent stack $\mathcal{X}$ regarded as a functor $\mathcal{X} : \mathbf{CAlg}_{[0, \infty)} \rightarrow \mathbf{Ani}$:

- It is a Kan extension along $\mathbf{CAlg}_{[0, \infty)}^{\text{ft}} \hookrightarrow \mathbf{CAlg}_{[0, \infty)}$.
- Its restriction to any $\mathbf{CAlg}_{[0, n]}$ is the Kan extension along $\mathbf{CAlg}_{[0, n]}^{\text{ft}} \hookrightarrow \mathbf{CAlg}_{[0, n]}$.

Notice that by definition $\mathbf{CAlg}_{[0, n]} \cap \mathbf{CAlg}_{[0, \infty)}^{\text{ft}} = \mathbf{CAlg}_{[0, n]}^{\text{ft}}$ □

todo 1.1.3.1. I've temporarily changed my mind. Maybe I should use cohomological indexing after all.

2. The geometric Satake equivalence

2.1. The affine Grassmannian

2.1.1. Basic constructions

Notation 2.1.1.1. Let k be a field, and $F = k((t))$ the local function field. Then let $\mathcal{O} = k[[t]]$ denote the ring of integers. We let $D_R = \text{Spec } R[[t]]$ and $D_R^* = \text{Spec } R((t))$ denote the (punctured) formal disc.

Definition 2.1.1.1. Let G be a smooth affine k -group. We let Gr_G , the affine Grassmannian of G be the functor of points given by $\text{Gr}_{G(R)} = \{(\mathcal{E}, \beta)\}$, where $\mathcal{E}$ is a G -torsor on D_R and $\beta : \mathcal{E}|_{D_R^*} \rightarrow \mathcal{E}^0|_{D_R^*}$ be a trivialization of $\mathcal{E}$ on the punctured disc (here $\mathcal{E}^0$ is the trivial bundle). We shall call the data β a modification of the bundle $\mathcal{E}$, sometimes denoted by $\beta : \mathcal{E} \dashrightarrow \mathcal{E}^0$

Note that this definition makes sense for general group schemes $\underline{G}$ over $\mathcal{O}$ or D_k (whereas in the above case $\underline{G} = G_{\mathcal{O}}$), which we also denote by $\text{Gr}_{\underline{G}}$.

Example 2.1.1.2. As a first example, take $G = \text{GL}_n$. Then $\text{Gr}_{G(R)}$ admits another functor of points description, given by the set of lattices $\Lambda \hookrightarrow R((t))^n$, where Λ is a projective $R[[t]]$ -module such that $\Lambda \otimes_{R[[t]]} R((t)) = R((t))^n$. To see that this is equivalent to the above definition, notice that a GL_n -principal bundle is equivalent to an n -dimensional vector bundle, and the projective module is the data for $\mathcal{E}$, and the second isomorphism is the trivialization away from 0.

Proposition 2.1.1.2. $\text{Gr}_{\underline{G}}$ is represented by an ind-scheme ind-of-finite-type over k . Moreover, if $\underline{G}$ is reductive, then $\text{Gr}_{\underline{G}}$ is ind-projective.

Proof: We only prove the case where G is constant, since the general fact relies on the fact that for any group $\underline{G}$ over $k[[t]]$, there is a faithful representation $\underline{G} \hookrightarrow \text{GL}_n$ for some n such that $\frac{\text{GL}_n}{\underline{G}}$ is quasi-affine, and when $\underline{G}$ is reductive, it is affine. In the constant case, we can obviously embed G into GL_n over k with the above properties and base change to $k[[t]]$. In this case, we have a locally closed embedding $\text{Gr}_G \hookrightarrow \text{Gr}_{\text{GL}_n}$ which is closed in the reductive case. It thus remains to show that Gr_{GL_n} is ind-projective.

To show this, we note that for every lattice $\Lambda \hookrightarrow R((t))^n$, there is some N such that $t^{-N}R[[t]]^n \hookrightarrow \Lambda \hookrightarrow t^N R[[t]]^n$. By definition, the functor of points for lattices Λ between $t^{-N}R[[t]]^n$ and $t^N R[[t]]^n$ is $\text{Gr}(2nN)$, the usual grassmannian. By the Plucker embedding, $\text{Gr}(2nN)$ is projective, and by definition $\text{Gr}_{\text{GL}_n} = \text{colim}_N \text{Gr}(2nN)$. Moreover, note that the transition maps are closed embeddings, so we have Gr_{GL_n} ind-projective. $\square$

We then give another characterization of the affine Grassmannian.

Definition 2.1.1.3. Let X denote a presheaf over $\mathcal{O}$. We define the n -th jet space of X $L^n X$ to be the presheaf over k such that $L^n X(R) = X\left(\frac{R[[t]]}{t^n}\right)$. Moreover, we define the positive loop space of X $L^+ X$ to be the presheaf over k such that $L^+ X(R) = X(R[[t]])$.

Then let X denote a presheaf over K . We define the loop space LX to be the presheaf over k such that $LX(R) = X(R((t)))$.

For X a presheaf over k , we write $L^n X$, $L^+ X$, LX short for $L^n(X \otimes_k \mathcal{O})$, $L^+(X \otimes_k \mathcal{O})$, and $L(X \otimes_k K)$.

Example 2.1.1.3. In particular, for $\underline{G}$ a group scheme over $\mathcal{O}$, we may define the loop group LG and the positive loop group L^+G , which would be of fundamental importance in the theory.

Theorem 2.1.1.4. $\mathrm{Gr}_{\underline{G}} = LG/L^+G$, where the right hand side is identified with the fpqc quotient stack.

Proof: We first construct a natural action $L\underline{G} \curvearrowright \mathrm{Gr}_{\underline{G}}$, given on R -points by $g \cdot (\mathcal{E}, \beta) = (\mathcal{E}, g\beta)$. Notice that we have a natural identification of $L\underline{G}(R)$ with the set of data $(\mathcal{E}, \beta, \varepsilon)$, where $(\mathcal{E}, \beta) \in \mathrm{Gr}_{\underline{G}}(R)$, and $\varepsilon : \mathcal{E} \rightarrow \mathcal{E}^0$ is a global trivialization. We have this isomorphism by mapping $g \in L\underline{G}(R)$ to $g \mapsto (\mathcal{E}^0, g, \mathrm{id})$ and the inverse $(\mathcal{E}, \beta, \varepsilon) \mapsto \beta \cdot \varepsilon$. Thus the projection $L\underline{G} \twoheadrightarrow \mathrm{Gr}_{\underline{G}}$ to the first two factors clearly identifies $L\underline{G}$ as a $L^+\underline{G}$ -torsor over $\mathrm{Gr}_{\underline{G}}$. $\square$

We now introduce the global version of the affine Grassmannian.

Definition 2.1.1.5. Let X be a reduced connected curve over k (not necessarily smooth), and let $x \in |X|$ be a closed point. Let $\underline{G}$ be a smooth affine group scheme over X , and we define $\mathrm{Gr}_{\underline{G}, x}$ to be the presheaf over k sending R to pairs $(\mathcal{E}, \beta)$, where $\mathcal{E}$ is a $\underline{G}$ -torsor on X_R , and $\beta : \mathcal{E}|_{X_R^*} \rightarrow \mathcal{E}^0|_{X_R^*}$ be a trivialization of $\mathcal{E}$ away from x (here X_R^* is the the base change of $X^* = X - x$ to R).

Definition 2.1.1.6. Let $x \in |X|$ be a closed point in a reduced connected curve over k . We let $\mathcal{O}_x$ denote the completed local ring at x (since we will not use the usual stalk in this section), and let $D_{x,R}$ denote $\mathrm{Spec} \mathcal{O}_x \hat{\otimes} R$.

Definition 2.1.1.7. Let $\underline{G}_x = \underline{G} \times_X x$. Moreover, let $\mathrm{res} : \mathrm{Gr}_{\underline{G}, x} \rightarrow \mathrm{Gr}_{\underline{G}_x}$ induced by restricting a $\underline{G}$ -torsor on X_R to $D_{x,R}$.

Proposition 2.1.1.8. The above restriction map is an isomorphism.

To prove [Proposition 2.1.1.8](#), we need the Beauville-Laszlo's descent theorem.

Theorem 2.1.1.9 (Beauville-Laszlo). Let $p : A \rightarrow \tilde{A}$ be a map of commutative rings. We fix a non-zero-divisor $f \in A$ and assume that $p(f)$ is also not a zero divisor. Moreover, we assume that the natural map $A/f \rightarrow \tilde{A}/p(f)$ is an isomorphism. Let $\mathbf{Mod}_A^f$ denote the f -torsion-free A -modules, and $\mathrm{Desc}(A, \tilde{A}, f)$ denote the category of the descent datum (M_1, M_2, φ) where $M_1 \in \mathbf{Mod}_{A[f^{-1}]}$, $M_2 \in \mathbf{Mod}_{\tilde{A}}^{p(f)}$, and $\varphi : M_1 \otimes_{A[f^{-1}]} \tilde{A}[p(f)^{-1}] \rightarrow M_2 \otimes_{\tilde{A}} \tilde{A}[p(f)^{-1}]$ is an isomorphism.

Now, by flatness of localization, we have $\varphi : M[f^{-1}] \otimes_{A[f^{-1}]} \tilde{A}[p(f)^{-1}] \simeq (M \otimes_A \tilde{A})[p(f)^{-1}]$, so $M \mapsto (M[f^{-1}], M \otimes_A \tilde{A}, \varphi)$ gives a functor $\mathbf{Mod}_A^f \rightarrow \mathrm{Desc}(A, \tilde{A}, f)$. This functor is then an equivalence, and finiteness, flatness, and finite projectivity are preserved under descent.

Proof: We now prove [Proposition 2.1.1.8](#). We directly construct an inverse to res , namely, given an $\underline{G}_x$ -torsor on $D_{x,R}$ with a trivialization β on $D_{x,R}^*$, we glue it with the trivial $\underline{G}$ -torsor on X_R^* by β via [Theorem 2.1.1.9](#) (we should find a vector bundle on X such that it is the associated bundle of our principal bundles, but I'm temporarily not quite sure on how to do this). $\square$

Corollary 2.1.1.10. The loop group LG_x represents the k -presheaf sending R to triples $(\mathcal{E}, \alpha, \beta)$, $\mathcal{E}$ is a G -torsor on X , α a trivialization of $\mathcal{E}|_{D_{x,R}}$, and β a trivialization of $\mathcal{E}|_{X_R^*}$.

remark 2.1.1.4. There is an alternate construction of loopgroups and affine Grassmannians over the mixed characteristic world, sometimes called the Witt Grassmannian. Namely, this time, $L^+G(R) = G(W(R))$, $LG(R) = G(W(R)[p^{-1}])$, $\text{Gr}_G = LG/L^+G$. They are sheaves over the perfect algebras over $\mathbb{F}_p$.

2.1.2. The affine Weyl group

Notation 2.1.2.1. From now on, we assume G is a reductive group over k . We fix a Borel subgroup $B \subset G$, and its maximal torus T . Denote the weight (character) lattice and coweight (cocharacter) lattice by $X^\bullet(T) = \text{Hom}(T, \mathbb{G}_m)$ and $X_\bullet(T) = \text{Hom}(\mathbb{G}_m, T)$ respectively. We let $\Phi \subset X^\bullet(T)$ denote the set of roots, and $\Phi^\vee \subset X_\bullet(T)$ denote the coroots. We denote the positive roots by Φ^+ , and the dominant coweights by $X_\bullet(T)^+$.

For $\mu \in X_\bullet(T)$ a coweight, we denote t^μ by the image of $t \in \mathbb{G}_m(K)$ by μ in $T(K)$.

Theorem 2.1.2.1. We have a natural isomorphism $G(\mathcal{O}) \backslash G(F) / G(\mathcal{O}) \simeq X_\bullet(T) / W = X_\bullet(T)^+$.

Proof: Recall that we have the Cartan decomposition by linear algebra

$$G(F) = \coprod_{\mu \in X_\bullet(T)} G(\mathcal{O}) t^\mu G(\mathcal{O})$$

which gives the desired decomposition. □

Construction 2.1.2.2. Let $\mathcal{E}_1$ and $\mathcal{E}_2$ be two trivial G -torsors over D with trivializations $\varphi_i : \mathcal{E}_i \rightarrow \mathcal{E}^0$, and let $\beta : \mathcal{E}_1|_{D^*} \rightarrow \mathcal{E}_2|_{D^*}$ be an isomorphism over D^* . Then, the composition $\varphi_2 \beta \varphi_1^{-1} \in \text{Aut}(\mathcal{E}^0|_{D^*}) = G(F)$, so obviously its image in $G(\mathcal{O}) \backslash G(F) / G(\mathcal{O}) = X_\bullet(T)^+$ does not depend on the trivializations φ_i , denoted by $\text{Inv}(\beta)$, the relative position of the modification β .

Now, given the Bruhat decomposition, as in the classical case, we may define the Schubert varieties. The Schubert varieties defined via the dominant coweights by the spherical Schubert varieties. But before we start, we need a lemma on group actions and orbits.

Lemma 2.1.2.3 (orbit lemma). Let G be a smooth group scheme and X an algebraic variety. Let $G \curvearrowright X$, then for any $x \in X$, the orbit $G \cdot x \subset X$ is locally closed in X .

We omit the proof, but the key of the proof is using that the image of maps of varieties is constructible, and finally using a dense open set to deduce locally closedness.

Definition 2.1.2.4. Recall $L^+G \curvearrowright LG$ hence on the quotient LG/L^+G (note that we take quotient on the right here, so the action is not trivial). We let Gr_μ denote the L^+G -orbit (naturally equipped with the reduced structure) of the image of the point $t^\mu \in LG \twoheadrightarrow \text{Gr}$. Then, we let

$$\text{Gr}_{\leq \mu} = \coprod_{\lambda \leq \mu} \text{Gr}_\lambda$$

We call $\text{Gr}_{\leq \mu}$ the spherical Schubert varieties and Gr_{μ} the spherical Schubert cell.

Alternatively, one may directly describe the locally closed subset Gr_{μ} via the modification.

Proposition 2.1.2.5. Gr_{μ} can also be identified as the subset of Gr

$$\text{Gr}_{\mu} = \{(\mathcal{E}, \beta) \in \text{Gr} \mid \text{Inv}(\beta) = \mu\}$$

Proof: Firstly, note that $t^{\mu} = (\mathcal{E}, \beta) \in \text{Gr}$ has obviously $\text{Inv}(\beta) = \mu$ by the definition of relative position and the projection $LG \rightarrow \text{Gr}$. Then, note that for $(\mathcal{E}', \beta') = g(\mathcal{E}, \beta)$ for some $g \in L^+G$, we have $\text{Inv}(\beta') = \text{Inv}(\beta)$ since the action of g on $(\mathcal{E}, \beta)$ is given by left multiplication on β , and in the definition of relative position, we have a quotient of $G(\mathcal{O})$ on the left. Thus, in a single L^+G -orbit, the relative position of β is always the same. Finally, by [Theorem 2.1.2.1](#), we have

$$\text{Gr} = \coprod (L^+G \backslash LG / L^+G) / L^+G = \coprod \text{Gr}_{\mu}$$

so each Gr_{μ} is exactly the orbit such that the invariant is equal to μ . □

remark 2.1.2.2. There is a slight subtlety in the above proof by confusing the set $G(F)$ ($G(\mathcal{O})$) with the ind-scheme LG (L^+G). However, since everything is reduced, and we are working over an algebraically closed field, no serious problem arises since any closed reduced subscheme over an algebraically closed field is determined by its closed points.

Proposition 2.1.2.6. Gr_{μ} is a quasi-projective variety of dimension $\langle 2\rho, \mu \rangle$. Moreover, we have $\text{Gr}_{\leq \mu}$ the Zariski closure of Gr_{μ} , so is a (not necessarily smooth) projective variety.

Proof: By definition, the stabilizer for $L^+G \curvearrowright \text{Gr}$ of t^{μ} is $L^+G / (L^+G \cap t^{\mu} L^+G t^{-\mu})$, which is firstly finite dimensional by calculating the dimension of its tangent space, which is given by

$$\mathfrak{g}(\mathcal{O}) / (\mathfrak{g}(\mathcal{O}) \cap \text{Ad}_{t^{\mu}} \mathfrak{g}(\mathcal{O})) = \bigoplus_{\langle \alpha, \mu \rangle \geq 0} \mathfrak{g}_{\alpha}(\mathcal{O}) / t^{\langle \alpha, \mu \rangle} \mathfrak{g}_{\alpha}$$

so $\dim \text{Gr}_{\mu} = \langle 2\rho, \mu \rangle$. Then, since it is finite dimensional locally closed subscheme of an ind-projective variety Gr , it is quasi-projective.

The second assertion is more difficult. We show that the Zariski closure of Gr_{μ} contains Gr_{λ} for any $\lambda \leq \mu$, and assume the spherical Schubert variety $\text{Gr}_{\leq \mu}$ is closed in Gr , so identifying $\text{Gr}_{\leq \mu}$ as the closure of Gr_{μ} . Then, recall that for $\lambda < \mu$, there is a sequence μ_i such that $\lambda = \mu_0 < \mu_1 < \dots < \mu_n = \mu$, where μ_i and μ_{i-1} differ by a positive simple root. Thus, it suffices to show that $t^{\mu-\alpha}$ lies in the closure of Gr_{μ} . To show this, we construct a curve C connecting the two points $t^{\mu-\alpha}$ □

Lemma 2.1.2.7. Θ is an effective Cartier divisor corresponding to the $\mathcal{O}(1)$ -bundle.

2.2. The proof of the Satake equivalence

2.2.1. The Satake category

2.2.2. The geometric Satake transform

Now that we have proven that the functor of taking hypercohomology $H^* : \text{Sat}_G \rightarrow \mathbf{Vect}_{\overline{\mathbb{Q}}_\ell}$ is a neutral Tannakian category, we may identify Sat_G with the representation category $\mathbf{Rep}_{\tilde{G}}$, where the Tannakian group $\tilde{G}$ is the algebraic group of automorphisms $\text{Aut}^\otimes H^*$. We now (and the proof of theorem suffices to) identify $\tilde{G}$ with the dual group of G over $\overline{\mathbb{Q}}_\ell$.

Definition 2.2.2.1. Recall a reductive group is completely determined by its root datum $(X_\bullet(T), \Phi^\vee, X^\bullet(T), \Phi)$. For a reductive group G over a algebraically closed field with its root datum given by $(X_\bullet(T), \Phi^\vee, X^\bullet(T), \Phi)$, its dual group $\hat{G}$ to be the reductive group with root datum $(X^\bullet(T), \Phi, X_\bullet(T), \Phi^\vee)$.

Notice that the base field of $\hat{G}$ can be arbitrary.

Example 2.2.2.1. GL_n has dual GL_n , and SL_n has dual PGL_n .

Lemma 2.2.2.2. $\tilde{G}$ is a connected reductive group over $\overline{\mathbb{Q}}_\ell$.

Proof: Firstly, we claim that the sheaves IC_μ for μ the fundamental coweights (the weights dual to the simple roots) form a set of tensor generators for Sat_G . Notice that the convolution Grassmannian $\text{Gr}_{\leq \mu} \tilde{\times} \text{Gr}_{\leq \nu} \rightarrow \text{Gr}_{\leq \mu+\nu}$ is birational. Now, given a proper birational map $f : X \rightarrow Y$, we prove IC_Y is a direct summand of $f_* \text{IC}_X$. Now, take an open set U contained in the smooth locus of Y such that f is an isomorphism on U . By definition, $f_* \text{IC}_X|_U = \text{IC}_Y|_U = k_U$. By the perverse continuation principle applied to the sheaves $f_* \text{IC}_X|_U$ and $\text{IC}_Y|_U$, we have a map $\text{IC}_Y \rightarrow f_* \text{IC}_X$ non-zero on U , so is non-zero. Thus since IC_Y is simple, it must be a subobject of $f_* \text{IC}_X$. Thus applied to our previous settings, $\text{IC}_{\mu+\nu}$ is a subobject hence a direct summand of the convolution sheaf $\text{IC}_\mu * \text{IC}_\nu$ by the decomposition theorem.

This provides that $\tilde{G}$ is connected and algebraic (i.e. the algebra $\mathcal{O}_{\tilde{G}}(\tilde{G})$ is of finite type) because $\bigotimes_\mu \text{IC}_\mu$ for μ running over fundamental coweights is a tensor generator for Sat_G . Then, since Sat_G is semisimple, we have furthermore that $\tilde{G}$ is reductive. $\square$

Theorem 2.2.2.3. Letting $(X^\bullet(T), \Phi^\vee, X_\bullet(T), \Phi)$ denote the root datum of G , we have the Tannakian group $\tilde{G}$ has root datum $(X_\bullet(T), \Phi, X^\bullet(T), \Phi^\vee)$, so $\tilde{G}$ can be identified with the dual group of G .

The proof of [Theorem 2.2.2.3](#) is by first proving the easy case where G is a torus, then reducing general cases to tori.

Now, in general, the affine Grassmannians can be very non-reduced, but since we are talking about ℓ -adic sheaves over the spaces, we may temporarily get rid ourselves of the nuisances by taking $(\text{Gr}_G)_{\text{red}}$. Thus, from now on, Gr_G will naturally mean the reduced versions without further notices.

Lemma 2.2.2.4. [Theorem 2.2.2.3](#) holds if $G = T$ is a torus.

Proof: We first discuss the structure of Gr_T . By definition, let $T = \mathbb{G}_m^n$, then $LT(R) = T(R((t))) = (R((t))^\times)^n$, and $L^+T(R) = (R[[t]]^\times)^n$. Then, by definition, we have $R((t))^\times/R[[t]]^\times$ identified by $\mathbb{Z}$, identified to the lowest exponent of t in each component. Thus this tuple of integers is given equivalently by a cocharacter $\lambda \in X_\bullet(T) \simeq \mathbb{Z}^n$. Therefore, Gr_T is a discrete set with points identified with $X_\bullet(T)$. Thus, Sat_T is the category of $X_\bullet(T)$ -graded finite dimensional vector spaces, and H^* is the functor forgetting the grading.

Now, since $X_\bullet(T)$ -graded vector spaces is tensor generated by 1_μ where μ are fundamental coweights, so an R -point of the Tannakian groups is given by maps $1_\mu \rightarrow R^\times$, hence by the tensor-automorphism condition, are given by $\text{Hom}_{\text{Ab}}(X_\bullet(T), R^\times)$. And by definition, this is the functor of points of the dual torus. $\square$

Now it suffices to reduce the case to $G = T$. To do this, we construct the categorical Satake transform

$$\text{CT} : \text{Sat}_G \rightarrow \text{Sat}_T$$

which is a symmetric monoidal functor from the Satake category of G to the one of its maximal torus T , which firstly fits in the following commutative diagram

$$\begin{array}{ccc} \text{Sat}_G & \xrightarrow{\text{CT}} & \text{Sat}_T \\ & \searrow H^* & \swarrow H^* \\ & \text{Vect}_{\overline{\mathbb{Q}}_\ell} & \end{array}$$

Construction 2.2.2.5. The main idea of constructing CT is via parabolic induction. We fix a borel subgroup $i : B \subset G$ and let $q : B \rightarrow T$ denote the projection to the abstract Cartan subgroup. The maps induces maps on the affine Grassmannians

$$\text{Gr}_T \xleftarrow{q} \text{Gr}_B \xrightarrow{i} \text{Gr}_G$$

Then, we have $\text{Gr}_T \simeq X_\bullet(T)$, and for $\lambda \in X_\bullet(T)$, we denote $S_\lambda = i(q^{-1}(\lambda)) \subset \text{Gr}_G$, which, by definition, can also be identified with $LU \cdot t^\lambda$, where $U \subset B$ is the unipotent subgroup. Moreover, we define

$$S_{\leq \lambda} = \bigcup_{\lambda' \leq \lambda} S_{\lambda'}$$

To construct and investigate the properties of the parabolic inductions, we start with some topological (and finally, cohomological) properties of S_λ .

Proposition 2.2.2.6. $S_{\leq \lambda}$ is closed with $S_\lambda \subset S_{\leq \lambda}$ an open dense subset.

Proof: We first show that $S_{\leq \lambda}$ is closed. We consider its moduli description. Let χ be a dominant integral weight, and V_χ denote the highest weight representation of G , and ℓ_χ denote its highest weight space. Then we have

$$S_{\leq \lambda} = \{(\mathcal{E}, \beta) \in \text{Gr} | \beta^{-1}(\ell_\chi) \subset t^{-(\chi, \lambda)}(V_{\chi, \mathcal{E}})\}$$

where $V_{\chi, \mathcal{E}}$ is the associated bundle of $\mathcal{E}$, namely, $\mathcal{E} \times^G V_\chi$, which is a vector bundle over D_k . Then, these conditions can be translated to a polynomial equation by passing to matrices, so is a closed subscheme of Gr_G .

Then, for the density of $S_\lambda \subset S_{\leq \lambda}$, it actually follows from the construction of the curve in [Proposition 2.1.2.6](#). □

Lemma 2.2.2.7. The set $S_{<\lambda}$ is the intersection of $\overline{S}_\lambda$ with a hyperplane section of the bundle $\mathcal{O}(1)$.

Proof: Let $2\rho^\vee$ denote the sum of positive roots of G . Then we choose some embedding $T \subset B$, and consider the map

$$\mathbb{G}_m \xrightarrow{2\rho^\vee} T \subset G \subset L^+G$$

Then the L^+G action on Gr_G restricts to a $\mathbb{G}_m$ -action on Gr_G via the above inclusion. Notice that the fixed points of this action is given by Gr_T by the Cartan decomposition, and we can identify the attractor of this action with Gr_B . □

Theorem 2.2.2.8. We have an isomorphism of functors

$$H^*(-) = \bigoplus_{\lambda} H_c^*(S_\lambda, -) : \mathrm{Sat}_G \rightarrow \mathbf{Vect}_{\overline{\mathbb{Q}}_\ell}$$

Proof:

□

3. The Bezrukavnikov-Finkelberg equivalence

We attempt to give a rather complete sketch of the infamous (at least among my friends) article of Bezrukavnikov-Finkelberg. The main theorem of the article is the following:

Theorem 3.1. $D_{\text{perf}}^{\check{G}}(U_h^{\square}) \simeq D_{L^+G \rtimes \mathbb{G}_m}(\text{Gr})$, and $D_{\text{perf}}^{\check{G}}(\text{Sym}^{\square}(\check{\mathfrak{g}})) \simeq D_{L^+G}(\text{Gr})$.

where the algebras $U_h^{\square}$ and $\text{Sym}^{\square}(\check{\mathfrak{g}})$ are to be defined later. The strategy of the proof is given by the following steps.

- (1) Construct functors $\kappa_h : \mathcal{HC}_h \rightarrow \text{QCoh}^{\mathbb{G}_m} \left(M_{(\check{\mathfrak{t}}/W)^2 \Delta} \right)$ and $\kappa : \text{Coh}^{\check{G} \times \mathbb{G}_m}(\check{\mathfrak{g}}^*) \rightarrow \text{Coh}^{\mathbb{G}_m} \left(T(\check{\mathfrak{t}}^*/W) \right)$.
The functors are fully faithful on the subcategory of free objects, namely $\mathcal{HC}_h^{\text{fr}}$ and $\text{Coh}_{\text{fr}}^{\check{G} \times \mathbb{G}_m}(\check{\mathfrak{g}}^*)$.
- (2) Extend the Satake functor $S : \mathbf{Rep}(\check{G}) \rightarrow \mathbf{Perv}_{L^+G \rtimes \mathbb{G}_m}(\text{Gr})$ to fully faithful functors

$$S_h : \mathcal{HC}_h^{\text{fr}} \rightarrow D_{L^+G \rtimes \mathbb{G}_m}(\text{Gr}), \text{ and } S_{qc} : \text{Coh}_{\text{fr}}^{\check{G} \times \mathbb{G}_m}(\check{\mathfrak{g}}^*) \rightarrow D_{L^+G}(\text{Gr})$$

These functors are constructed via the fully faithful cohomology functors

$$H_{L^+G \rtimes \mathbb{G}_m}^* : \mathbf{Perv}_{L^+G \rtimes \mathbb{G}_m}(\text{Gr}) \rightarrow \text{Coh}^{\mathbb{G}_m} \left(M_{(\check{\mathfrak{t}}/W)^2 \Delta} \right), H_{L^+G}^* : \mathbf{Perv}_{L^+G}(\text{Gr}) \rightarrow \text{Coh}^{\mathbb{G}_m} \left(T(\check{\mathfrak{t}}^*/W) \right)$$

and by analysing the images, we have $\kappa_h|_{\mathcal{HC}_h^{\text{fr}}}$ and $\kappa|_{\text{Coh}_{\text{fr}}^{\check{G} \times \mathbb{G}_m}(\check{\mathfrak{g}}^*)}$ factors through the cohomology functors by S_h and S_{qc} , i.e., $\kappa_h = H_{L^+G \rtimes \mathbb{G}_m}^* \circ S_h$ and $\kappa = H_{L^+G}^* \circ S_{qc}$.

- (3) Extend S_h and S_{qc} to our final equivalences via the fully faithful inclusions $\mathcal{HC}_h^{\text{fr}} \subset D_{\text{perf}}^{\check{G}}(U_h^{\square})$ and $\text{Coh}_{\text{fr}}^{\check{G} \times \mathbb{G}_m}(\check{\mathfrak{g}}^*) \subset D_{\text{perf}}^{\check{G}}(\text{Sym}^{\square}(\check{\mathfrak{g}}))$.

We will expand the three steps one by one.

3.1. Step 1

3.1.1. Computations of equivariant cohomology

Before we start, we must compute the equivariant cohomologies of points.

Definition 3.1.1.1. Let X be a scheme and $Z \subset X$ be a closed subscheme classified by the quasi-coherent sheaf of ideals $\mathcal{I}$. We construct the deformation to the normal cone as following. Firstly, take $\mathbb{A}^1 = \text{Spec } k[\hbar]$. Then, consider the following extended Rees $\mathcal{O}_X$ -algebra $\mathcal{R}$ given by the sheaf of subalgebras in $\mathcal{O}_X[\hbar^{\pm}]$ generated by $\hbar$ and $f\hbar^{-1}$ where $f \in \mathcal{I}$. We define the deformation to the normal cone $M_X Z = \text{Spec}_X \mathcal{R}$.

Definition 3.1.1.2. For groups G with $H_G^*(\text{pt})$ is an integral domain, we define the localized equivariant cohomology $H_G^*(X)_{\text{loc}} = H_G^*(X) \otimes_{H_G^*(\text{pt})} \text{Frac}(H_G^*(\text{pt}))$.

Theorem 3.1.1.3. There is an equivalence

$$\alpha : \mathcal{O} \left(N_{(\check{\mathfrak{t}}/W)^2 \Delta} \right) \simeq \bigoplus_{\pi_1(G)} H_{L^+G \rtimes \mathbb{G}_m}^*(\text{Gr})$$

Here Δ denote the diagonal embedding.

Proof: We first treat the case where G is simply connected.

Firstly, notice $H_{L^+G \rtimes \mathbb{G}_m}^*(\mathrm{Gr}) = H_{\mathbb{G}_m}^*(L^+G \backslash LG / L^+G)$ admit a bimodule structure of $H_{L^+G}^*(\mathrm{pt})$. Since we have the homotopy equivalence $L^+G \simeq G$ by contractibility of $k[[t]]$, we have $H_{L^+G}^*(\mathrm{pt}) = H_G^*(\mathrm{pt}) = \mathcal{O}(\mathfrak{t}/W)$ by Chern-Weil theory. Thus, we have $H_{L^+G \rtimes \mathbb{G}_m}^*(\mathrm{Gr})$ is a $H_{L^+G}^*(\mathrm{pt}) \otimes H_{L^+G}^*(\mathrm{pt}) \otimes H_{\mathbb{G}_m}^*(\mathrm{pt}) = \mathcal{O}((\mathfrak{t}/W)^2 \times \mathbb{A}^1)$. This gives a structure map

$$\alpha : \mathcal{O}((\mathfrak{t}/W)^2 \times \mathbb{A}^1) \rightarrow H_{L^+G \rtimes \mathbb{G}_m}^*(\mathrm{Gr})$$

Then we show it extends to the extended Rees algebra $\mathcal{R}$, that is, we show that for $f \in \mathcal{I}_\Delta$, we have $\alpha(f)$ contains a factor of $\alpha(\hbar)$ where we regard $H_{\mathbb{G}_m}^*(\mathrm{pt}) = k[\hbar]$, i.e., the two maps $H_{L^+G}^*(\mathrm{pt}) \rightarrow H_{L^+G \rtimes \mathbb{G}_m}^*(\mathrm{Gr})/\hbar$ coincide. This is because $H_{L^+G \rtimes \mathbb{G}_m}^*(\mathrm{Gr})/\hbar = H^*(L^+G \backslash LG / L^+G)$. Thus we obtain a map

$$\alpha : \mathcal{O}(M_{(\mathfrak{t}/W)^2} \Delta) \rightarrow H_{L^+G \rtimes \mathbb{G}_m}^*(\mathrm{Gr})$$

We then show it is an equivalence. For injectivity, note $H_{L^+G \rtimes \mathbb{G}_m}^*(\mathrm{pt}) = \mathcal{O}(\mathfrak{t}/W \times \mathbb{A}^1)$. Notice by definition that $\mathcal{O}(M_{(\mathfrak{t}/W)^2} \Delta)$ is an integral domain, so it suffices to prove the localized morphism

$$\alpha_{\mathrm{loc}} : \mathcal{O}(M_{(\mathfrak{t}/W)^2} \Delta) \otimes_{\mathcal{O}(\mathfrak{t}/W \times \mathbb{A}^1)} \mathrm{Frac}(\mathcal{O}(\mathfrak{t} \times \mathbb{A}^1)) \rightarrow H_{L^+G \rtimes \mathbb{G}_m}^*(\mathrm{Gr}) \otimes_{\mathcal{O}(\mathfrak{t}/W \times \mathbb{A}^1)} \mathrm{Frac}(\mathcal{O}(\mathfrak{t} \times \mathbb{A}^1))$$

is an injection. This is because the right hand side can be identified with the localized equivariant cohomology $H_{T \times \mathbb{G}_m}^*(\mathrm{Gr})_{\mathrm{loc}}$. $\square$

3.1.2. The involved categories

Definition 3.1.2.1. Let $U = U(\check{\mathfrak{g}})$, and let $U_{\hbar}$ denote the graded enveloping algebra, explicitly, the graded $k[\hbar]$ -algebra generated by $\check{\mathfrak{g}}$ under the relations $xy - yx = \hbar[x, y]$. Equivalently, $U_{\hbar}$ is the Rees algebra of the filtered algebra U .

These enveloping algebras are equipped with the structure of $\check{G}$ -representations via the adjoint action $\check{G} \curvearrowright \check{\mathfrak{g}}$.

Definition 3.1.2.2. We define the category of $\hbar$ -Harish-Chandra bimodules $\widetilde{\mathcal{HC}}_{\hbar}$ to be the category with the following objects with the requirements:

- (1) An object M is a graded $U_{\hbar} \otimes_{k[\hbar]} U_{\hbar}$ -module equipped with the structure of a $\check{G}$ -representation $\rho : \check{G} \curvearrowright M$, such that:
- (2) the action map $U_{\hbar} \otimes_{k[\hbar]} U_{\hbar} \otimes M \rightarrow M$ is $\check{G}$ -equivariant, and,
- (3) for $x \in \check{\mathfrak{g}}$, the action of $1 \otimes x + x \otimes 1$ on M is $\hbar \cdot d\rho(x)$.

The intuition to this definition is some sort of Harish-Chandra modules of Harish-Chandra pairs.

We define $\mathcal{HC}_{\hbar} \subset \widetilde{\mathcal{HC}}_{\hbar}$ to be the full subcategory spanned by the modules M such that M is finitely generated as $U_{\hbar} \otimes 1$ -modules.

4. Koszul duality

4.1. A first glance: Koszul duality for finite type groups

4.1.1. Highest weight categories

In this section, we introduce a special type of Abelian categories which behaves like the category of perverse sheaves over a suitable stratified space or the BGG category $\mathcal{O}$. The history of the

developments of such categories is complicated, resulting in various definitions of such categories (some of which are simple generalizations of others). We will begin with the first definition.

Definition 4.1.1.1. Let $\mathcal{A}$ be an Abelian category over some field k . We say $\mathcal{A}$ is highest weight in the sense of [CPS88] if the following conditions are met.

- (1) $\mathcal{A}$ satisfies the condition AB5 and is locally Artinian. That is, every object is generated by finite-length objects by filtered colimits. In particular, every object admit a (possibly infinite) composition series.
- (2) There is an interval finite poset Λ (that is, the intervals $[\mu, \nu] = \{\lambda \mid \mu \leq \lambda \leq \nu\} \subset \Lambda$ are finite) with the set of all non-isomorphic simple objects $\text{IC}_\lambda \in \mathcal{A}$ indexed by Λ .
- (3) There is a collection of costandard objects ∇_λ indexed by Λ .
- (4) These objects contain $\text{IC}_\lambda \subset \nabla_\lambda$ as a subobject, and the quotient $\nabla_\lambda / \text{IC}_\lambda$ contains only IC_μ with $\mu > \lambda$ in the composition series. Moreover, $[\nabla_\mu : \text{IC}_\nu]$ and $\dim_k \text{Hom}(\nabla_\mu, \nabla_\nu)$ are finite.
- (5) IC_λ admit an injective envelope I_λ in $\mathcal{A}$. This injective I_λ admit another complete filtration

$$0 = F_0 \subset F_1 \subset \cdots \subset I_\lambda$$

such that $F_1 = \nabla_\lambda$, and $F_i / F_{i-1} = \nabla_\mu$ where $\mu > \lambda$ for $i > 1$. Moreover, each ∇_μ appears only finitely many times.

This is not the main focus of this section, so we only present the main theorem of the article.

Theorem 4.1.1.2 (Theorem 3.9, [CPS88]). Finite length highest weight categories with finitely many simple objects are always of the form as a right module category $\mathbf{Mod}_A^{f.g.}$ for some right quasi-hereditary algebra A .

This is mainly used in the paper [BGS96]. In their paper, they only focused on finite length categories with finitely many simple objects, namely, the category of perverse sheaves with respect to a nice enough finite stratification. Their definitions went as follows.

Definition 4.1.1.3. Let $\mathcal{A}$ be an Abelian category over some field k . We say $\mathcal{A}$ is highest weight in the sense of [BGS96] if the following conditions hold.

- (1) $\mathcal{A}$ is of finite length.
- (2) $\mathcal{A}$ has simple objects indexed by a finite poset Λ , denoted by IC_λ . Moreover, $\text{End}(\text{IC}_\lambda) = k$. The poset Λ will be referred to as the weight poset.
- (3) There are standard and costandard objects Δ_λ and ∇_λ indexed by Λ , equipped with structure morphisms $\Delta_\lambda \rightarrow \text{IC}_\lambda$ and $\text{IC}_\lambda \rightarrow \nabla_\lambda$.
- (4) Let $\Lambda' \subset \Lambda$ be any subposet closed under taking smaller elements. We define $\mathcal{A}_{\Lambda'}$ to be the subcategory of $\mathcal{A}$ spanned by objects with simple components $\text{IC}_{\lambda'}$ for $\lambda' \in \Lambda'$. Then, for any maximal element $m \in \Lambda'$, we have $\Delta_m \rightarrow \text{IC}_m$ a projective cover and $\text{IC}_m \rightarrow \nabla_m$ an injective hull in $\mathcal{A}_{\Lambda'}$.
- (5) $\ker(\Delta_\lambda \rightarrow \text{IC}_\lambda)$ and $\text{coker}(\text{IC}_\lambda \rightarrow \nabla_\lambda)$ lie in $\mathcal{A}_{<\lambda}$.
- (6) $\text{Ext}^2(\Delta_\mu, \nabla_\nu) = 0$.

Lemma 4.1.1.4. We have the following orthogonality and Ext vanishing conditions:

$$\dim_k \operatorname{Hom}(\Delta_\mu, \nabla_\nu) = \delta_{\mu, \nu}, \operatorname{Ext}^i(\Delta_\mu, \nabla_\nu) = 0$$

Proof: For the first orthogonality condition, note $\Delta_\mu \rightarrow \operatorname{IC}_\mu$ is a projective cover in $\mathcal{A}_{\leq \mu}$, so every quotient of Δ_μ must have IC_μ as a factor. Similarly, every subobject of ∇_ν must intersect IC_ν . Thus, if there is a non-trivial map $\Delta_\mu \rightarrow \nabla_\nu$, since the image and coimage is non-zero, we have

$$[\Delta_\mu : \operatorname{IC}_\nu] \neq 0, [\Delta_\nu : \operatorname{IC}_\mu] \neq 0$$

which implies $\nu \leq \mu$ and $\mu \leq \nu$, so $\mu = \nu$. Then, to compute $\dim_k \operatorname{Hom}(\Delta_\mu, \nabla_\mu)$, by the fifth axiom, we have

$$\operatorname{Hom}(\Delta_\mu, \nabla_\mu) = \operatorname{End}(\operatorname{IC}_\mu) = k$$

For the Ext^1 vanishing, we put $\Lambda' = \{\leq \mu\} \cup \{\leq \nu\}$. Then, either μ or ν or both is maximal in Λ' , so the exact sequence

$$0 \rightarrow \nabla_\nu \rightarrow M \rightarrow \Delta_\mu \rightarrow 0$$

splits by either projectivity of ∇_ν or injectivity of Δ_μ in $\mathcal{A}_{\Lambda'}$.

For higher Ext vanishing, we prove by descending induction on μ . If μ is maximal in Λ , then Δ_μ is projective and there is nothing to prove. In general, consider $P_\mu \rightarrow \operatorname{IC}_\mu$, which has a standard filtration with Δ_μ the last term. Thus considering the long exact sequence of $\operatorname{Hom}(-, \nabla_\nu)$ applied to

$$0 \rightarrow K \rightarrow P_\mu \rightarrow \Delta_\mu \rightarrow 0$$

we have

$$\operatorname{Ext}^{i-1}(K, \nabla_\nu) \rightarrow \operatorname{Ext}^i(\Delta_\mu, \nabla_\nu) \rightarrow \operatorname{Ext}^i(P_\mu, \nabla_\nu)$$

exact with the first term and last term 0 by induction. Thus the claim follows. $\square$

Proposition 4.1.1.5. Such categories have enough projective objects, with each projective admitting a finite filtration with standard associated graded pieces. The dual argument applies for injective objects.

Proof: We prove the following statement by induction: There are enough projectives in $\mathcal{A}_{\Lambda'}$, and each one admit a standard filtration.

We prove by induction on the cardinality of Λ' . The case where $\#\Lambda' = 1$ is trivial. Then, suppose Λ' and $\Lambda' \cup \{\lambda\}$ are both closed under taking smaller elements. We will prove the statement for $\Lambda' \cup \{\lambda\}$ from the statement for Λ' .

Now, for any $\mu \in \Lambda' \cup \{\lambda\}$, we attempt to construct a projective cover of IC_μ in $\mathcal{A}_{\Lambda' \cup \{\lambda\}}$. Since Λ' is closed under taking smaller elements, it is necessary that λ is an maximal element. Thus, $\Delta_\lambda \rightarrow \operatorname{IC}_\lambda$ is a desired projective cover. Then, for $\mu \in \Lambda'$, we wish to upgrade the projective cover $P \rightarrow \operatorname{IC}_\mu$ in $\mathcal{A}_{\Lambda'}$ to a projective cover in $\mathcal{A}_{\Lambda' \cup \{\lambda\}}$. We put $E = \operatorname{Ext}^1(P, \Delta_\lambda)$. Since P is not yet projective, E somehow measures the extend that P is non-projective. Then consider the unit element in

$$1 \in E \otimes E^* = \operatorname{Ext}^1(P, E^* \otimes \Delta_\lambda)$$

This gives rise to an extension

$$0 \longrightarrow E^* \otimes \Delta_\lambda \longrightarrow \tilde{P} \longrightarrow P \longrightarrow 0$$

such that the composition

$$E \rightarrow \text{Hom}(E^* \otimes \Delta_\lambda, \Delta_\lambda) \xrightarrow{\delta} \text{Ext}^1(P, \Delta_\lambda)$$

is the identity. Then, since the first map is an isomorphism because $\text{End}(\Delta_\lambda) = \text{End}(\text{IC}_\lambda) = k$, the boundary map is also an isomorphism. Then, by projectivity of Δ_λ , $\text{Ext}^1(\Delta_\lambda, \Delta_\lambda) = 0$.

Hence, the long exact sequence

$$\text{Hom}(E^* \otimes \Delta_\lambda, \Delta_\lambda) \longrightarrow \text{Ext}^1(P, \Delta_\lambda) \longrightarrow \text{Ext}^1(\tilde{P}, \Delta_\lambda) \longrightarrow \text{Ext}^1(E^* \otimes \Delta_\lambda, \Delta_\lambda)$$

gives $\text{Ext}^1(\tilde{P}, \Delta_\lambda) = 0$. Now, it suffices to prove that $\tilde{P}$ is a projective cover for IC_μ . This is equivalent to saying

$$\dim_k \text{Hom}(\tilde{P}, \text{IC}_\nu) = \delta_{\mu, \nu}, \text{Ext}^1(\tilde{P}, \text{IC}_\nu) = 0$$

where the second condition ensures that $\tilde{P}$ is projective and the first condition ensures that $\tilde{P} \rightarrow \text{IC}_\mu$ is a cover. Consider the exact sequence

$$\text{Ext}^1(P, \text{IC}_\nu) \longrightarrow \text{Ext}^1(\tilde{P}, \text{IC}_\nu) \longrightarrow \text{Ext}^1(E^* \otimes \Delta_\lambda, \text{IC}_\nu)$$

the first term vanishes by the projectivity of P in $\mathcal{A}_{\Lambda'}$, and the third vanishes by projectivity of Δ_λ . So the second group vanishes. By the same reason, $\dim_k \text{Hom}(\tilde{P}, \text{IC}_\nu) = \delta_{\mu, \nu}$. Thus, it suffices to prove the statement for IC_λ . Then, to compute $\text{Hom}(\tilde{P}, \text{IC}_\lambda)$, consider the exact sequence

$$0 \longrightarrow K \longrightarrow \Delta_\lambda \longrightarrow \text{IC}_\lambda \longrightarrow 0$$

Then, $\text{Hom}(\tilde{P}, \text{IC}_\lambda) = 0$ because K lies in $\mathcal{A}_{\Lambda'}$ and run the exact sequence, and Ext^1 vanishes by the vanishing condition of Ext^2 .

Hence, the original statement follows from induction. $\square$

Corollary 4.1.1.6. BGG reciprocity holds, that is, we have $(P_\mu : \Delta_\nu) = [\nabla_\nu : L_\mu]$

Proof: By the Hom orthogonality, the first number is equal to $\dim_k \text{Hom}(P_\mu, \nabla_\nu)$. For the second, consider the Jordan-Holder series of ∇_ν

$$0 = F_0 \subset \cdots \subset F_n = \nabla_\nu$$

By the projectivity of P_μ , we have

$$0 \longrightarrow \text{Hom}(P_\mu, F_{i-1}) \longrightarrow \text{Hom}(P_\mu, F_i) \longrightarrow \text{Hom}(P_\mu, \text{IC}_\lambda) \longrightarrow 0$$

exact, where IC_λ is a simple factor. But then, since $P_\mu \rightarrow \text{IC}_\mu$ is a projective cover, $P_\mu \rightarrow \text{IC}_\lambda$ is nontrivial iff $\lambda = \mu$, so the second number is equal to $\dim_k \text{Hom}(P_\mu, \nabla_\nu)$. $\square$

Proposition 4.1.1.7. Let $X = \coprod_{\lambda \in \Lambda} X_\lambda$ be an affine stratification. Then, $\mathbf{Perv}_\Lambda(X)$ is a highest weight category in the sense of [BGS96].

Proof: We verify the axioms. The first three axioms are obtained directly by the properties of perverse sheaves on an affine stratified space. For the fourth property, it suffices to prove that $*$ - or $!$ -restriction to the open strata is exact (which is standard), because

$$\mathrm{Hom}(\Delta, -) = \mathrm{Hom}(j_! 1, -) = \mathrm{Hom}(1, j^*(-)) = j^*(-)$$

and the dual holds for ∇ . Moreover, these objects are indecomposable, so they are the projective cover and injective hull. The fifth property also follows from the property of intermediate extensions. The final property is slightly more difficult, and is proven by the following property of perverse sheaves: firstly, by an analysis of extensions we have

$$\mathrm{Ext}_{\mathbf{Perv}_\Lambda}^2(\Delta_\mu, \nabla_\nu) \subset \mathrm{Hom}_{D_\Lambda}^2(\Delta_\mu, \nabla_\nu)$$

and for the latter, we have

$$\begin{aligned} \mathrm{Hom}_{D_\Lambda}^2(\Delta_\mu, \nabla_\nu) &= \mathrm{Hom}_{D_\Lambda}(j_{\mu!} 1_\mu[\dim X_\mu], j_{\nu*} 1_\nu[\dim X_\nu + 2]) \\ &= \mathrm{Hom}_{D_\Lambda}(j_\nu^* j_{\mu!} 1_\mu[\dim X_\mu], 1_\nu[\dim X_\nu + 2]) \end{aligned}$$

Then, the first term is 0 for $\nu \neq \mu$, and if $\nu = \mu$, then the above Hom is an ext group of vector spaces, which is 0. $\square$

Corollary 4.1.1.8. We have the realization functor an equivalence.

$$D(\mathbf{Perv}_\Lambda(X)) \xrightarrow{\simeq} D_\Lambda(X)$$

Proof: Note that P_λ and ∇_λ separately generates both $D(\mathbf{Perv}_\Lambda(X))$ and $D_\Lambda(X)$. It suffices to show that

$$\mathrm{Ext}_{\mathbf{Perv}_\Lambda(X)}^i(P_\mu, \nabla_\nu) = \mathrm{Hom}_{D_\Lambda(X)}^i(P_\mu, \nabla_\nu)$$

By projectivity of P_μ , the left hand side vanishes unless $i = 0$. The right side also vanishes unless $i = 0$ by the standard filtration of P and applying the property of $!$ -pushforwards. The statement for $i = 0$ is trivial. $\square$

Now, we may wish to generalize the above properties to possibly infinite stratifications. This is where the modern definition of highest weight categories come in.

Definition 4.1.1.9. Let $\mathcal{A}$ be an Abelian category over some field k . We say $\mathcal{A}$ is highest weight in the sense of [Ric17] if it is highest weight in the sense of [BGS96], but with the condition of the finite poset Λ replaced by for any $\lambda \in \Lambda$, the set $\{\mu | \mu \leq \lambda\} \subset \Lambda$ is finite.

Properties such as the orthogonality and Ext vanishing still holds. However, the category of perverse sheaves over a stratified space may not be highest weight, for example

$$D(I \setminus LG/I) \neq D(\mathbf{Perv}(I \setminus LG/I))$$

which is an evidence of $\mathbf{Perv}(I \setminus LG/I)$ not being highest weight. Later on, we will adopt this definition of highest weight categories.

Lemma 4.1.1.10. Let $\Lambda' \subset \Lambda$ be a subposet closed under taking smaller objects, and let $\mathcal{A}$ be a highest weight category with weight poset Λ . Then, the Serre quotient $\pi : \mathcal{A} \twoheadrightarrow \mathcal{A}/\mathcal{A}_{\Lambda'}$ is a highest

weight category with weight poset $\Lambda - \Lambda'$. Its simple, standard, and costandard objects are the image of the projection π of the original simple, standard, and costandard objects.

Proof: Axioms (1), (2), (4), (5) are obvious. Then, for (4), we claim that we have an isomorphism

$$\mathrm{Hom}_{\mathcal{A}}(\Delta_\lambda, N) = \mathrm{Hom}_{\mathcal{A}/\mathcal{A}_{\Lambda'}}(\pi(\Delta_\lambda), \pi(N))$$

for $\lambda \in \Lambda - \Lambda'$. This is because a map $\pi(\Delta_\lambda) \rightarrow \pi(N)$ is represented by a map $M' \rightarrow N/N'$ in $\mathcal{A}$, where $M' \subset \Delta_\lambda$ and $N' \subset N$ such that Δ_λ/M and N' is in $\mathcal{A}_{\Lambda'}$. Then, since every quotient of Δ_λ necessarily contains $\mathrm{IC}_\lambda \notin \mathcal{A}_{\Lambda'}$, we have $M' = \Delta_\lambda$. Then, since $N' \in \mathcal{A}_{\Lambda'} \subset \mathcal{A}_{\leq \lambda}$, and that Δ_λ is projective in $\mathcal{A}_{\leq \lambda}$, we have $\mathrm{Ext}^1(\Delta_\lambda, N') = 0$, which implies that the map $\Delta_\lambda \rightarrow N/N'$ factors through $N \rightarrow N/N'$. Hence, we have the above map is surjective. Moreover since $M' = \Delta_\lambda$ necessarily, and any map $\Delta_\lambda \rightarrow N/N'$ does not annihilate IC_λ , it is non-zero. Thus the above map is an isomorphism. This implies that Δ_λ is projective in $(\mathcal{A}/\mathcal{A}_{\Lambda'})_{\Lambda''}$ for λ maximal in Λ'' .

Finally for axiom (6), it suffices to prove the case where Λ is finite because to compute $\mathrm{Ext}^2(\pi(\Delta_\mu), \pi(\nabla_\nu))$, it suffices to compute this in the category $\mathcal{A}_{\leq \mu, \nu}$, which is finite by assumption. However, in the finite case, we have enough projectives, which allows us to prove the claim as follows. We consider the object $\pi(P_\mu) \twoheadrightarrow \pi(\Delta_\mu)$. The kernel of this map has a filtration by $\pi(\Delta_\lambda)$ where $\lambda > \mu$. Then, if $\pi(P_\mu)$ is projective, the above vanishing of Ext^2 follows by the long exact sequence given by

$$\mathrm{Ext}^1(K, \pi(\nabla_\nu)) \longrightarrow \mathrm{Ext}^2(\pi(\Delta_\mu), \pi(\nabla_\nu)) \longrightarrow \mathrm{Ext}^2(\pi(P_\mu), \pi(\nabla_\nu))$$

where the last term is 0 by projectivity and the first is 0 by analysing extensions. Thus it suffices to prove projectivity. If not, consider a non-split surjection $\pi(M) \twoheadrightarrow \pi(P_\mu)$. This surjection is represented by a map $M' \rightarrow P_\mu/N'$ where M/M' and N' lies in $\mathcal{A}_{\Lambda'}$. Moreover, by surjectivity, this map must have cokernel lying in $\mathcal{A}_{\Lambda'}$. However, since $P_\mu \twoheadrightarrow \mathrm{IC}_\mu$ is a projective cover with $\mu \notin \Lambda'$, the cokernel must necessarily vanish, so the representing map is itself surjective. Thus, $M' \twoheadrightarrow P_\mu$ splits, which contradicts $\pi(M) \twoheadrightarrow \pi(P_\mu)$ not split. $\square$

Proposition 4.1.1.11. Let $\mathcal{A}$ be a highest weight category with weight poset Λ . Then we have the following.

- (1) The embedding $\iota : D(\mathcal{A}_{\Lambda'}) \hookrightarrow D(\mathcal{A})$ is fully faithful.
- (2) The embedding $\iota : D(\mathcal{A}_{\Lambda'}) \hookrightarrow D(\mathcal{A})$ admit left and right adjoints, denoted by ι^L and ι^R .
Moreover, the projection $\pi : D(\mathcal{A}) \rightarrow D(\mathcal{A}/\mathcal{A}_{\Lambda'})$ admit left and right adjoints, denoted by π^L and π^R . These adjoints provide a semi-orthogonal decomposition of $D(\mathcal{A})$, namely

Definition 4.1.1.12. Tilting objects

todo 4.1.1.1. Finish this section.

4.1.2. Where is Koszul duality?

4.2. Koszul Duality for Kac-Moody groups

4.2.1. Kac-Moody lie algebras and Kac-Moody groups

Definition 4.2.1.1. We define a generalized Cartan matrix of size ℓ to be a matrix $A \in \text{Mat}_{\ell \times \ell}(\mathbb{Z})$, such that the following conditions are satisfied:

- (1) $a_{i,i} = 2$;
- (2) $a_{i,j} \leq 0$ for $i \neq j$;
- (3) $a_{i,j} = 0$ iff $a_{j,i} = 0$.

We define a realization of a generalized Cartan matrix to be a triple $(\mathfrak{h}, \pi, \pi^\vee)$, where $\mathfrak{h}$ is a complex $(2\ell - \text{rank } A)$ -dimensional vector space, $\pi = \{\alpha_1, \dots, \alpha_\ell\} \subset \mathfrak{h}^*$ a set of roots, and $\pi^\vee = \{\alpha_1^\vee, \dots, \alpha_\ell^\vee\} \subset \mathfrak{h}$ a set of coroots, subject to the relation $\alpha_j(\alpha_i^\vee) = a_{i,j}$.

By linear algebra, such realization exists and is unique up to isomorphism.

Construction 4.2.1.2. Given a generalized Cartan matrix A , we construct its associated Kac-Moody Lie algebra $\mathfrak{g}(A)$ by the following presentation:

- It is generated by $\mathfrak{h}$ and symbols $\{e_1, \dots, e_\ell, f_1, \dots, f_\ell\}$.
- The relations are given by
 - (1) $[\mathfrak{h}, \mathfrak{h}] = 0$.
 - (2) $[h, e_i] = \alpha_i(h)e_i$, $[h, f_i] = -\alpha_i(h)f_i$.
 - (3) $[e_i, f_j] = \delta_{i,j}\alpha_i^\vee$.
 - (4) $(\text{ad } e_i)^{1-a_{i,j}}(e_j) = 0$ for $i \neq j$, and $(\text{ad } f_i)^{1-a_{i,j}}(f_j) = 0$.

We denote $\mathfrak{n}$ and $\mathfrak{n}^-$ by the subalgebra generated by $\{e_i\}$ and $\{f_i\}$ respectively. We define $\mathfrak{b} = \mathfrak{n} + \mathfrak{h}$ and $\mathfrak{b}^- = \mathfrak{n}^- + \mathfrak{h}$ the standard Borel subalgebras. We also define $\tilde{\mathfrak{g}}$ to be the Lie algebra generated by the same generators but only subject to the relations (1), (2), and (3).

Definition 4.2.1.3. Let $\alpha \in \mathfrak{h}^*$. We define the root space $\mathfrak{g}_\alpha \subset \mathfrak{g}$ to be the subspace

$$\mathfrak{g}_\alpha = \{x \in \mathfrak{g} \mid [h, x] = \alpha(h)x, \forall h \in \mathfrak{h}\}$$

We define $Q = \bigoplus \mathbb{Z}\alpha_i$, and $Q^+ = \bigoplus_{\geq 0} \mathbb{Z}\alpha_i$ the root lattice and nonnegative root lattice. Q^- is defined likewise. We say $\alpha \in Q^+$ is a positive root if $\mathfrak{g}_\alpha \neq 0$. We denote the set of positive roots by R^+ . We likewise define R and R^- .

For a root α , we define $|\alpha| = \sum n_i$, where $\alpha = \sum n_i \alpha_i$.

Theorem 4.2.1.4. We have the following decompositions for $\mathfrak{g}$:

- (1) $\mathfrak{g} = \mathfrak{n} \oplus \mathfrak{h} \oplus \mathfrak{n}^-$ as vector spaces.

$$(2) \quad \mathfrak{n}^\pm = \bigoplus_{\alpha \in Q^\pm} \mathfrak{g}_\alpha$$

Moreover, each root space is finite dimensional.

Proof: Omitted. □

Construction 4.2.1.5. Then, we slightly enlarge the above constructed Lie algebra. We define the complete version of the Lie algebra

$$\hat{\mathfrak{n}} = \prod_{\alpha \in R^+} \mathfrak{g}_\alpha$$

equipped with the natural Lie bracket. This is well-defined because each root space is finite dimensional. Then, notice the family of ideals

$$\hat{\mathfrak{n}}(k) = \prod_{\substack{\alpha \in R^+ \\ |\alpha| \geq k}} \mathfrak{g}_\alpha$$

gives $\hat{\mathfrak{n}}$ a structure of a pro-Lie algebra. Moreover, it is pro-nilpotent. Thus, its exponential gives a well-defined pro-algebraic group $\mathcal{U}$, namely given by

$$\mathcal{U} = \lim_k \exp(\hat{\mathfrak{n}}/\hat{\mathfrak{n}}(k))$$

4.2.2. A realization result

Definition 4.2.2.1. Consider the category $\mathbf{Vect}_{\overline{\mathbb{Q}}_\ell}$. We define a Frobenius module M to be an object $M \in \mathbf{Mod}_{\overline{\mathbb{Q}}_\ell}$ equipped with an automorphism $\mathrm{Fr}_M : M \rightarrow M$. We denote the category of Frobenius modules by $\mathbf{Mod}(\mathrm{Fr})$. We define a Frobenius module to be locally finite if it is a union of its finite submodules. For any Frobenius module M , we denote M^f by the union of its finite submodules.

There is clearly a forgetful functor $\mathbf{Mod}(\mathrm{Fr}) \rightarrow \mathbf{Vect}_{\overline{\mathbb{Q}}_\ell}$ given by $M \mapsto M^{\mathrm{Fr}}$, the fixed point functor. Later on, we will use this functor to realize a category enriched over $\mathbf{Mod}(\mathrm{Fr})$ to a DG category over $\overline{\mathbb{Q}}_\ell$.

Definition 4.2.2.2. We define a locally finite Frobenius module to be continuous if the action $\mathbb{Z} \curvearrowright M$ through Frobenius extends to an action $\widehat{\mathbb{Z}} \curvearrowright M$. For a locally finite Frobenius module, we may define its weights as in the usual convention. For a general Frobenius module M , we make the above definitions after applying M^f .

From now on, we assume every Frobenius module to be continuous with integral weights.

Definition 4.2.2.3. Let M be a Frobenius module. We define $M^{\mathrm{Fr}\text{-unip}}$ to be the Frobenius submodule of M on which Fr_M acts unipotently. We define M^{op} to denote the Frobenius module with the same underlying vector space as M , but with the inverse Frobenius action.

Definition 4.2.2.4. Let E be a $\overline{\mathbb{Q}}_\ell$ -algebra. Let E be equipped with a continuous Frobenius module structure compatible with its algebra structure, we define $\mathbf{Mod}_{E, \mathrm{Fr}}$ to be the category of E -modules equipped with a Frobenius automorphism compatible with the Frobenius module structure on E .

In this section, our main goal is to answer the following question.

Question 4.2.2.5. For a category $\mathcal{D}$ enriched in $\mathbf{Mod}_{\mathrm{Fr}}$, when do we have a fully faithful inclusion $\mathcal{D} \hookrightarrow \mathbf{Mod}_{E, \mathrm{Fr}}$?

We will show this with the following assumptions.

Assumption 4.2.2.6. Let $\mathcal{D}$ to be a $\overline{\mathbb{Q}}_\ell$ -linear category enriched in $D(\mathrm{Fr})$ (equivalently, the Hom and Ext groups are Frobenius modules), and is equipped with the following data:

- (1) A finite poset Λ such that $\mathcal{D}$ has a recollement indexed by Λ (that is, a series of open-closed semi-orthogonal decompositions). We define $\mathcal{D}_\lambda = \mathcal{D}_{\leq \lambda} / \mathcal{D}_{< \lambda}$.
- (2) A full additive subcategory $\mathcal{C}_\lambda \subset \mathcal{D}_\lambda$ stable under taking tensor with unipotent Frobenius modules, such that for any objects $c_1, c_2 \in \mathcal{C}_\lambda$, we have

$$\mathrm{Ext}_{\mathcal{D}_\lambda}^i(c_1, c_2)^{\mathrm{Fr}\text{-unip}} = 0$$

- (3) $\mathcal{C}_\lambda$ generates $\mathcal{D}_\lambda$ as a triangulated category. Moreover, there is an object $\mathbb{1}_\lambda \in \mathcal{C}_\lambda$ such that every object in $\mathcal{C}_\lambda$ is of the form $\mathbb{1}_\lambda \otimes M$ for some $M \in \mathbf{Mod}_{\mathrm{Fr}}$ (however, for arbitrary M , the object $\mathbb{1}_\lambda \otimes M \in \mathcal{D}_\lambda$ may not lie in $\mathcal{C}_\lambda$).
- (4) There exists an object $c_\lambda \in \mathcal{C}_{\leq \lambda}$ such that $i_\lambda^* c_\lambda = \mathbb{1}_\lambda$. Moreover, the induced ring morphism

$$i_\lambda^* : \mathrm{End}(c_\lambda) \rightarrow \mathrm{End}(\mathbb{1}_\lambda)$$

has a nilpotent kernel.

We let $\mathcal{C}$ denote the full subcategory of $\mathcal{D}$ consisting of objects d such that $i_\lambda^* d, i_\lambda^! d \in \mathcal{C}_\lambda$.

In practice, $\mathcal{D}$ will be a category of mixed ℓ -adic sheaves constructible with respect to a certain stratification, and $\mathcal{C}_\alpha$ the weight 0 constant sheaves (normally we would have locally constant, but we require constant in our case) over a strata. Then $\mathcal{C}$ will be the category of weight 0 sheaves.

The main theorem is the following, which would be our main technical tool later on.

Theorem 4.2.2.7. For a category $\mathcal{D}$ satisfying the requirements in [Assumption 4.2.2.6](#), there is a $\overline{\mathbb{Q}}_\ell$ -algebra E with a Frobenius action, such that $\mathcal{D} \hookrightarrow D(E, \mathrm{Fr})$ is fully faithful in the sense that both categories are enriched in $\mathbf{Set}$ rather than $D(\mathrm{Fr})$.

We will begin with a simple lemma on the computation of a homotopy fixed point.

Lemma 4.2.2.8. Let $M \in D(\mathrm{Fr})$. Then,

$$0 \longrightarrow H^{i-1}(M)_{\mathrm{Fr}} \longrightarrow H^i(M^{h\mathrm{Fr}}) \longrightarrow H^i(M)^{\mathrm{Fr}} \longrightarrow 0$$

Proof: Consider the following description of homotopy fixed points of a $\mathbb{Z}$ action as a fiber

$$M^{h\mathrm{Fr}} \longrightarrow M \xrightarrow{\mathrm{Fr} - \mathrm{id}} M$$

Then the desired exact sequence follows from the long exact sequence of the fiber sequence. $\square$

Corollary 4.2.2.9. Let $\mathcal{D}$ satisfy the condition (1) and (2) in [Assumption 4.2.2.6](#). Then, for $c_1, c_2 \in \mathcal{C}$, we have

$$\mathrm{Ext}_{\mathcal{D}}^i(c_1, c_2)^{\mathrm{Fr-unip}} = 0, i \neq 0$$

and

$$H^i(R\mathrm{Hom}_{\mathcal{D}}(c_1, c_2)^{h\mathrm{Fr}}) = \begin{cases} \mathrm{Hom}_{\mathcal{C}}(c_1, c_2)^{\mathrm{Fr}} & \text{if } i = 0 \\ \mathrm{Hom}_{\mathcal{C}}(c_1, c_2)_{\mathrm{Fr}} & \text{if } i = 1 \\ 0 & \text{otherwise} \end{cases}$$

Proof: For the first assertion, we apply induction on the stratas. On one strata, this is precisely condition (2). Then consider the excision triangle, which gives rise to the following long exact sequence for $c_1, c_2 \in \mathcal{C}_{\leq \lambda}$

$$\cdots \longrightarrow \mathrm{Ext}_{\mathcal{C}_{< \lambda}}^i(i_{< \lambda}^* c_1, i_{< \lambda}^! c_2)^{\mathrm{Fr-unip}} \longrightarrow \mathrm{Ext}^i(c_1, c_2)^{\mathrm{Fr-unip}} \longrightarrow \mathrm{Ext}_{\mathcal{C}_{\lambda}}^i(i_{\lambda}^* c_1, i_{\lambda}^* c_2)^{\mathrm{Fr-unip}} \longrightarrow \cdots$$

which gives the vanishing result.

For the second assertion, notice that for any $M \in \mathbf{Mod}_{\mathrm{Fr}}$, we have $M^{\mathrm{Fr}} \subset M^{\mathrm{Fr-unip}}$. The rest follows from the previous lemma. $\square$

Lemma 4.2.2.10. We adopt the decreasing filtration convention. Then, there is an equivalence of categories

$$\widetilde{\mathrm{gr}} : \mathrm{Fil}(\mathcal{C}) \rightarrow \mathrm{Ch}(\mathcal{C})$$

where $\mathrm{Fil}(\mathcal{C})$ is the full subcategory of $\mathrm{Fil}(\mathcal{D})$ such that $\mathrm{gr}^i(F^\bullet d) \in \mathcal{C}[-i]$, and $\mathrm{Ch}(\mathcal{C})$ is the category of chain complexes without any homotopy relations. Here, chain complexes mean the composite of two differentials are homotopic to 0. The equivalence is given by

$$\widetilde{\mathrm{gr}}(F^\bullet d) = \cdots \longrightarrow \mathrm{gr}^i(d)[i] \longrightarrow \mathrm{gr}^{i+1}(d)[i+1] \longrightarrow \cdots$$

where the transition maps are given by the following triangle

$$\mathrm{gr}^{i+1}(F^\bullet d) \longrightarrow \mathrm{gr}^{[i, i+1]}(F^\bullet d) \longrightarrow \mathrm{gr}^i(F^\bullet d) \longrightarrow \mathrm{gr}^{i+1}(F^\bullet d)[1]$$

Proof: This is well-known if $\mathcal{C}$ is a stable ∞ -category. Now, $\mathcal{C}$ is a additive subcategory of such, the proof is likewise. $\square$

Definition 4.2.2.11. Let $K_{\mathrm{acyc}}(\mathcal{C}) \subset K(\mathcal{C})$ denote the subcategory of the bounded homotopy category spanned by objects equivalent to $0 \in \mathcal{D}$.

Lemma 4.2.2.12. For any $c \in \mathcal{C}$, the functors

$$\mathrm{Hom}(c, -)^{\mathrm{Fr-unip}}, \mathrm{Hom}(-, c)^{\mathrm{Fr-unip}} : K_{\mathrm{acyc}}(\mathcal{C}) \rightarrow K_{\mathrm{acyc}}(\mathcal{C})$$

maps any complex to an exact sequence.

Proof: Fix an object $\mathcal{F} \in \mathrm{Fil}(\mathcal{C})$ such that $\widetilde{\mathrm{gr}} \mathcal{F} \in K_{\mathrm{acyc}}(\mathcal{C})$. Consider the spectral sequence

$$E_1^{p,q} = \mathrm{Ext}^{p+q}(c, \mathrm{gr}^p \mathcal{F}) = \mathrm{Ext}^q(c, \mathrm{gr}^p \mathcal{F}[p]) \implies \mathrm{Ext}^{p+q}(c, \mathcal{F})$$

However, since $\tilde{\mathcal{F}} \in K_{\text{acyc}}(\mathcal{C})$, $\mathcal{F} \simeq 0 \in \mathcal{D}$. Thus, the above spectral sequence converges to 0. Then, applying $(-)^{\text{Fr-unip}}$ to it, the spectral sequence degenerates at page 1 because q is forced to be 0. Thus, since the spectral sequence also converges to zero, the complex at $q = 0$ is exact. $\square$

Definition 4.2.2.13. We define $\mathcal{C}^{\text{Fr-unip}}$ to be the category with the same objects as $\mathcal{C}$, but with Hom groups replaced by

$$\text{Hom}_{\mathcal{C}^{\text{Fr-unip}}}(c_1, c_2) = \text{Hom}_{\mathcal{C}}(c_1, c_2)^{\text{Fr-unip}}$$

Construction 4.2.2.14. Notice that the forgetful functor $\mathcal{C} \rightarrow \mathcal{C}^{\text{Fr-unip}}$ admits a pro-left adjoint $\mathcal{C}^{\text{Fr-unip}} \rightarrow \text{Pro}(\mathcal{C})$, given by $c \mapsto c[[t]] = \lim c[t]/t^n = c \otimes \overline{\mathbb{Q}_\ell}[[t]]$, where Frobenius acts on $\overline{\mathbb{Q}_\ell}[[t]]$ by multiplying by $\exp t$. Explicitly, we have

$$\text{Hom}(c, c')^{\text{Fr-unip}} = \text{Hom}(c[[t]], c')^{\text{Fr}}$$

We remark that the exponential here is used to ensure unipotency.

Proposition 4.2.2.15. We have a well-defined functor $K(\mathcal{C})/K_{\text{acyc}}(\mathcal{C}) \rightarrow \mathcal{D}$. Moreover, it is fully faithful, and is essentially surjective if $\mathcal{C}_\alpha$ generates $\mathcal{D}_\alpha$ as a triangulated category.

Proof: Recall that maps $\text{Hom}_{K(\mathcal{C})/K_{\text{acyc}}(\mathcal{C})}(c_1, c_2)$ are computed by spans $c_1 \leftarrow c \rightarrow c_2$, where $\text{cofib}(c \rightarrow c_1) \in K_{\text{acyc}}(\mathcal{C})$. We may assume that $c_1 \in \mathcal{C}$, and we first show that taking the two term complex $c_1[[t]] \rightarrow c_1[[t]]$ is an initial (pro-) object mapping to c_1 with an acyclic cone. For this, we suppose $c \rightarrow c_1$ has an acyclic cone. Writing

$$c = \cdots \rightarrow c^{-1} \rightarrow c^0 \rightarrow c^1 \rightarrow \cdots$$

the cone is of the form

$$\cdots \rightarrow c^{-1} \rightarrow c^0 \rightarrow c^1 \oplus c_1 \rightarrow \cdots$$

Thus, by [Lemma 4.2.2.12](#), taking $\text{Hom}(c_1, -)^{\text{Fr-unip}}$, we obtain a long exact sequence

$$\cdots \rightarrow \text{Hom}(c_1, c^0)^{\text{Fr-unip}} \rightarrow \text{Hom}(c, c^1 \oplus c_1)^{\text{Fr-unip}} \rightarrow \cdots$$

This amounts to the exact sequence

$$\cdots \rightarrow \text{Hom}(c_1[[t]], c^0)^{\text{Fr}} \rightarrow \text{Hom}(c_1[[t]], c^1 \oplus c_1)^{\text{Fr}} \rightarrow \cdots$$

Note the Frobenius equivariant map $(0, p) : c_1[[t]] \rightarrow c^1 \oplus c_1$ has zero image, thus it comes from a map $\varphi_0 : c_1[[t]] \rightarrow c^0$. Then, since we want $- \times t : c_1[[t]] \rightarrow c_1[[t]]$, we also have $t\varphi_0 : c_1[[t]] \rightarrow c^0$ zero, so it comes from a map $\varphi^{-1} : c_1[[t]] \rightarrow c^{-1}$. Thus, we obtain a commutative square

$$\begin{array}{ccc} c_1[[t]] & \xrightarrow{t} & c_1[[t]] \\ \varphi_{-1} \downarrow & & \downarrow \varphi_0 \\ c^{-1} & \longrightarrow & c^0 \end{array}$$

which in turn exhibits initialness.

Hence, $\mathrm{Hom}_{K(c)/K_{\mathrm{acyc}}(c)}^\bullet(c_1, c_2)^{h\mathrm{Fr}}$ is equivalently the two term complex

$$0 \longrightarrow \mathrm{Hom}(c_1[[t]], c_2)^{\mathrm{Fr}} \xrightarrow{(\times t)^*} \mathrm{Hom}(c_1[[t]], c_2)^{\mathrm{Fr}} \longrightarrow 0$$

which, by the universal property, is the complex

$$0 \longrightarrow \mathrm{Hom}(c_1, c_2)^{\mathrm{Fr-unip}} \xrightarrow{\log \mathrm{Fr}} \mathrm{Hom}(c_1, c_2)^{\mathrm{Fr-unip}} \longrightarrow 0$$

which is precisely the complex computing $\mathrm{Hom}_{\mathcal{D}}^\bullet(c_1, c_2)^{\mathrm{Fr}}$, by [Corollary 4.2.2.9](#). Thus the functor is fully faithful. The image is clearly generated by the $\mathcal{C}_\alpha$'s, in a somewhat inaccurate sense. $\square$

Now, we notice that $\mathcal{C}^{\mathrm{Fr-unip}}$ is in some sense the Frobenius semisimplification, as opposed to $\mathcal{C}^{\mathrm{Fr}}$ is removing the Frobenius structure. This can be shown in the following lemma.

Lemma 4.2.2.16. From now on, we require $\mathcal{D}$ to satisfy all assumptions from [Assumption 4.2.2.6](#). For any $c \in \mathcal{C}$, there are complexes $M_\lambda \in D(\mathrm{Fr})$, such that

$$c = \bigoplus c_\lambda \otimes M_\lambda \in \mathcal{C}^{\mathrm{Fr-unip}}$$

Proof: Notice that we have $\mathrm{Ext}_{\mathcal{C}_{<\lambda}}^1(i_{<\lambda}^* c_1, i_{<\lambda}^! c_2)^{\mathrm{Fr-unip}} = 0$, so the restriction map

$$\mathrm{Hom}_{\mathcal{C}}(c_1, c_2)^{\mathrm{Fr-unip}} \twoheadrightarrow \mathrm{Hom}_{\mathcal{C}_\lambda}(i_\lambda^* c_1, i_\lambda^* c_2)^{\mathrm{Fr-unip}}$$

is surjective.

To prove the statement, we apply induction on the support of c . Suppose $c \in \mathcal{C}_{\leq \lambda}$. Then we have $i_\lambda^* c = \mathbb{1}_\lambda \otimes M_\lambda$. Moreover, since the left isomorphism is naturally Frobenius invariant, by the above surjection, the isomorphism lifts to a retraction diagram

$$c_\lambda \otimes M_\lambda \rightarrow c \rightarrow c_\lambda \otimes M_\lambda$$

This is a retraction since $\mathrm{End}(c_\lambda) \rightarrow \mathrm{End}(\mathbb{1}_\lambda)$ has nilpotent kernel, so any preimage of an isomorphism is an isomorphism.

Since retractions splits off, we may assume $c = c_\lambda \otimes M_\lambda \oplus c'$, where c' is supported on $\mathcal{C}_{<\lambda}$. Thus by induction, the proof is finished. $\square$

Now, suppose there is a family of objects d_λ also pulling back to $\mathbb{1}_\lambda$. For example, $d_\lambda = c_\lambda$ is a valid choice.

Lemma 4.2.2.17. $\mathcal{D}$ is generated by $\{d_\lambda\}_{\lambda \in \Lambda}$.

Proof: Again we apply induction on the stratas. Suppose $\mathcal{D}_{<\lambda}$ is generated by d_μ s for $\mu < \lambda$. Then, by adjunction, we have a map $i_{\lambda!} \mathbb{1}_\lambda \rightarrow d_\lambda$, whose cofiber lies in $\mathcal{D}_{<\lambda}$, so is generated by d_μ s for $\mu < \lambda$. Thus, $i_{\lambda!} \mathbb{1}$ is generated by d_μ s for $\mu \leq \lambda$. But $\mathcal{D}_{\leq \lambda}$ is generated by $i_{\lambda!} \mathbb{1}_\lambda$ and $\mathcal{D}_{<\lambda}$ by assumption (3). Thus, we are done. $\square$

Proof: Now we are ready to prove [Theorem 4.2.2.7](#). We let $d = \bigoplus d_\lambda$, and take the $\overline{\mathbb{Q}_\ell}$ -algebra E to be

$$E = \bigoplus_{i \in \mathbb{Z}} \mathrm{Ext}_{\mathcal{D}}^i(d, d)^{\mathrm{op}}$$

In particular, this is concentrated at degree 0, so not a DG algebra. We will prove that we have a fully faithful functor $\mathcal{C} \hookrightarrow \mathbf{Mod}_{E, \mathrm{Fr}}$, and then take derived categories, and prove

$K(\mathcal{C})/K_{\text{acyc}}(\mathcal{C}) \hookrightarrow D(E, \text{Fr})$ is fully faithful. Finally, by [Proposition 4.2.2.15](#), we obtain the desired fully faithful map.

Firstly, the functor $\mathcal{C} \hookrightarrow \mathbf{Mod}_{E, \text{Fr}}$ is via the obvious Yoneda functor

$$h = \bigoplus_{i \in \mathbb{Z}} \text{Ext}_{\mathcal{D}}^i(d, -) : \mathcal{C} \rightarrow \mathbf{Mod}_{E, \text{Fr}}$$

Note that by definition

$$\text{Hom}_{\mathbf{Mod}_{E, \text{Fr}}}(h(c_1), h(c_2)) = \text{Hom}_E(h(c_1), h(c_2))^{\text{Fr}}$$

so we only need

$$\text{Hom}_{\mathcal{C}}(c_1, c_2)^{\text{Fr-unip}} \xrightarrow{\simeq} \text{Hom}_E(h(c_1), h(c_2))^{\text{Fr-unip}}$$

For this, if we have $c_1 = d$, then

$$\text{Hom}_E(h(d), h(c_2))^{\text{Fr-unip}} = \text{Hom}_E(E, h(c_2))^{\text{Fr-unip}} = h(c_2)^{\text{Fr-unip}} = \text{Hom}_{\mathcal{C}}(d, c_2)^{\text{Fr-unip}}$$

And then, by projecting to direct summands in $\mathcal{C}^{\text{Fr-unip}}$, we see the above map is an isomorphism for d_λ s. Finally, since the d_λ s generate $\mathcal{C}^{\text{Fr-unip}}$ under direct sums, we have the functor fully faithful.

Then passing to homotopy categories, we first prove $K_{\text{acyc}}(\mathcal{C})$ is null-homotopic in $K(\mathcal{C}^{\text{Fr-unip}})$. This is done by [Lemma 4.2.2.12](#), and explicitly constructing the contracting homotopy via the exactness of Hom functors. Thus, we have a well-defined map $K(\mathcal{C})/K_{\text{acyc}}(\mathcal{C}) \rightarrow D(E, \text{Fr})$.

However, in this case, the Ext groups are computed likewise as above, so we are done. $\square$

Finally, we will also be able to compute the image of the above fully faithful functor.

Corollary 4.2.2.18. The image of the functor $\mathcal{D} \hookrightarrow \mathbf{Mod}(E, \text{Fr})$ is the full subcategory generated by the twists of $\text{Hom}(d, d_\lambda)$ s as a triangulated category.

Proof: Obvious. $\square$

5. The Bezrukavnikov-Arkhipov equivalence

See my thesis.

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